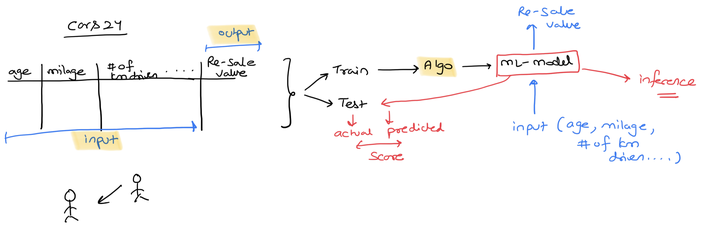
Cars 24



Recap

LLM ← Traditional chatbot
Attention is all you need

Model → [weights or parameters]

GPT
Sonnet
: → # of trained parameters
↓
accuracy & advancement of the model

3.5
↓
4o → Agent
↓
Image

Data

Text
1) Internet
2) Book
3) Research paper
4) Reddit
5) fB post / Tweets

→ Bias
→ garbage

→ quality of Data
→ Computational power

clean data
1) Removing duplicates
2) Bias
3) misinformation
4) wrong english → social media english
5) Abusive and threats

code files
knowledge
+
English (grammar)
→ Text / Book data

quantum mech.
Basic english grammar

[The capital of India is New Delhi]

Input
→ The
→ The capital
→ The capital of
→ The capital of India
→ The capital of India is
→ The capital of India is new

output
capital of India is new Delhi ✓
of India is new Delhi ✓
India is new Delhi ✓
is new Delhi ✓
new Delhi
Delhi

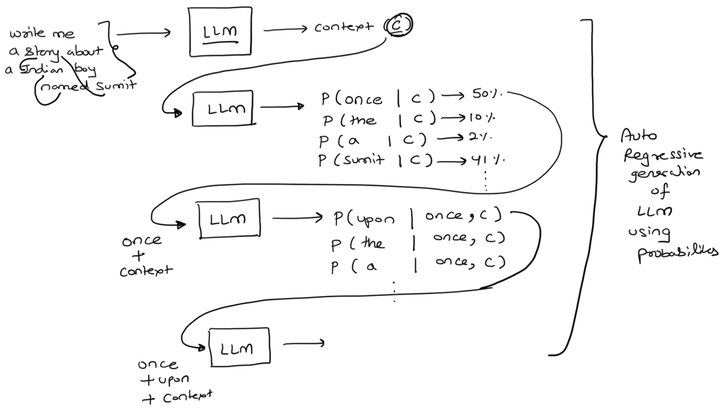
Tokenizer
Vocab Dict
embedding model

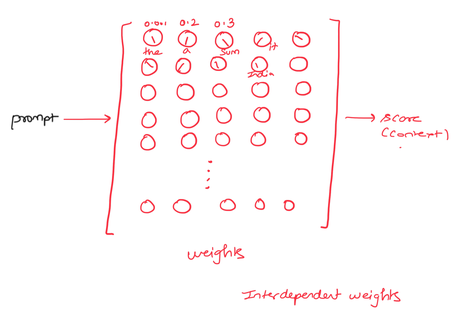
Token id	embedding
3825	[ ]

→ Tokenization → Tokenizers
→ embedding

Mathematical machine
Algo → Transformers

(weights)
write me a story of an Indian boy named Sumit
→ LLM → once upon a time....





Tokenization

Sequence of words → Set of numbers
So that the algo can distinguish them

Sequence of words → Tokenizers → [389, 276, 851, 357]
They were trained on set of data using BPE (Byte pair encoding)
to map most common seq. of characters to a particular ID

Sumit → not a common english word
Sum
it
Sumit

My name is Sumit



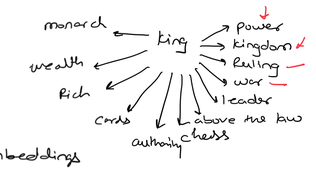
3825	my
1308	name
382	is
4215	Sum
278	it

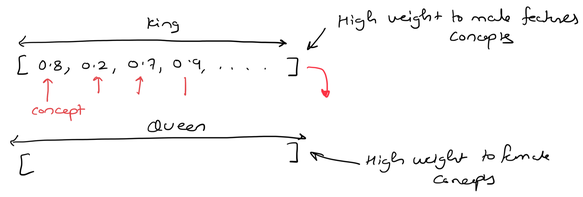
→ Embeddings

Algo (not be sufficient)
(not carry any meaning to them)
they are meaningless

[0.8, 0.2, 0.9 . . . .] meaning

[Transformers]

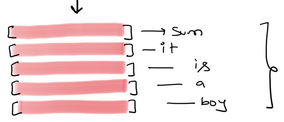




king → Embedding model → [_, _, _, _, _, _]

[king − man + woman ≈ queen]

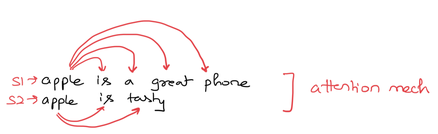
[knowledge preparation]
English Data → cleaned → [Input → output] Sumit is a boy
[Tokenizer → embedding]



[Transformer]

Each word is trying to understand their role in association with other words in the sente

attention mech



attention mech

[Context]